

Computer Algebra (SCI19 3112) — Final Cheat Sheet

Groups/Euler/crypto · Complex/polar/roots · Iso · FTA/fields · Coeff/value · FFT/IFFT · GCD/systems

1 — Groups Recap + Euler φ

Group: closure, identity e (unique, $aoe=a$), associativity, inverses (unique). $(\mathbb{Z}, +)$ infinite group; $(\mathbb{Z}_n, +)$ finite, always cyclic (gen $[1]$); $\mathbb{Z}_n^*$ group $\iff n$ prime. $|G|=\#\text{elem}$; $\text{ord}(g)$ = smallest $n>0$ with $g^n=e$.

Lagrange: $\text{ord}(g) \mid |G|$, $g^{|G|}=e$

Cyclic: $G=\langle g \rangle \iff \text{ord}(g)=|G|$

Worked: $|\mathbb{Z}_8^+|=8$; $\text{ord}[1]=8$; $\text{ord}[2]=4$ (2, 4, 6, 0); $\text{ord}[7]=8$ (7, 6, 5, 4, 3, 2, 1, 0); gens $[1], [3], [5], [7]$. $\varphi(n)=\#\{1 \leq k \leq n : \gcd(k,n)=1\}$. **Formulas:** p prime $\varphi(p)=p-1$; $\varphi(p^k)=p^k-p^{k-1}$; coprime $\varphi(mn)=\varphi(m)\varphi(n)$.

Worked: $\varphi(16)=8$ (1, 3, 5, 7, 9, 11, 13, 15).

$\varphi(17)=16$

$\varphi(14)=\varphi(2)\varphi(7)=(2-1)(7-1)=1 \cdot 6=6$

$\varphi(9)=9-3=6$ (prime power)

Euler: $a^{\varphi(n)} \equiv 1 \pmod{n}$ if $\gcd(a,n)=1$ (= Lagrange on units mod n , $|G|=\varphi(n)$). FLT = prime case ($\varphi(p)=p-1$) = Lagrange on $\mathbb{Z}_p^*$. **Use:** prime $\rightarrow$ FLT (smaller exponent); composite $\rightarrow$ Euler. **Idea:** reduce exponent mod $\varphi(n)$.

Worked $3^{347} \bmod 14$: Given $\gcd(3,14)=1$, $\varphi(14)=6$.

Step 1 (reduce exponent): $347=6 \cdot 57+5$ ($6 \cdot 57=342$).

Step 2 (evaluate): $3^5=243$;

$243-17 \cdot 14=243-238=5 \Rightarrow [5]$. **Check:** $\text{pow}=5$.

2 — Crypto

Caesar: $c \equiv m+s \pmod{26}$, $m \equiv c-s$. **Worked** BY ($B=1, Y=24$), $s=20$: $1+20=21=V$, $24+20=44-26=18=S \Rightarrow \text{VS}$.

Power: $c \equiv m^b \pmod{p}$; $m \equiv c^d$, $bd \equiv 1 \pmod{p-1}$. **Worked** $p=29, b=5$: $B: 1^5=1=B$.

Y Step 1 ($24 \equiv -5$): $(-5)^2=25$; **Step 2:**

$25^2=625$; $625-21 \cdot 29=625-609=16$.

Y Step 3: $16 \cdot (-5)=-80$;

$-80+3 \cdot 29=-80+87=7 \Rightarrow \text{H}$. d (test d or Euclid):

$5d \equiv 1 \pmod{28} \Rightarrow d=17$ ($5 \cdot 17=85=3 \cdot 28+1$).

Correctness = bijective: $D(E(x))=x$, $E(D(y))=y$. Caesar always bijective (shift). Power bijective $\iff \gcd(b, p-1)=1$ (else many-to-one). Many-to-one: ≥ 2 symbols map together; surjective: covers whole target.

Check decrypt: VS-20: $21-20=1=B$,

$18-20=-2+26=24=Y$. Power: $7^{17} \bmod 29=24=Y$

Check OK.

3 — Complex Numbers

Rect $z=a+bi$ ($\text{Re}=a$, $\text{Im}=b$), plane vector (a, b) .

Polar $r(\cos \theta + i \sin \theta)$, $r=|z|=\sqrt{a^2+b^2}$. $|z|^2=z\bar{z}$, $\bar{z}=a-bi$.

$+$, $-$ = vector + (parallelogram). Mult:

$(a+bi)(c+di)=(ac-bd)+(ad+bc)i$.

Worked 3+4i: $|z|=5$,

$z^{-1}=(3-4i)/25=0.12-0.16i$.

Worked $(1+6i)/(1-6i)$:

$(1+6i)^2/37=(-35+12i)/37$.

$\approx -0.9459+0.3243i$ **Check** $(37 \cdot (-0.9459)=-35$, $37 \cdot 0.3243=12)$.

Euler: $e^{i\theta}=\cos \theta + i \sin \theta$; **DeMoivre:**

$(re^{i\theta})^n=r^n(\cos n\theta + i \sin n\theta)$.

n -th roots: $\sqrt[n]{r}(\cos(\theta+2k\pi)/n + i \sin(\theta+2k\pi)/n)$, $k=0..n-1$.

Div: $z_1/z_2=z_1\bar{z}_2/|z_2|^2$; **Inverse:** $z^{-1}=\bar{z}/|z|^2$.

Polar mult: $r_1 r_2 e^{i(\theta_1+\theta_2)}$;

$e^{i\pi/3} \cdot e^{i\pi/6}=e^{i\pi/2}=i$.

Worked $8e^{i\pi/3}$: $8(\frac{1}{2}+i\frac{\sqrt{3}}{2})=4+4\sqrt{3}i \approx 4+6.9282i$.

Worked $11e^{3\pi i/2}$: $11(0-i)=-11i$.

Cube roots of $e^{i\pi/2}$: angles

$\pi/6, 5\pi/6, 3\pi/2 \Rightarrow \sqrt{3}/2+i/2, -\sqrt{3}/2+i/2, -i$.

Square roots of $e^{i\pi/4}$: angles $\pi/8, 9\pi/8$:

$\cos \frac{\pi}{8}=\sqrt{2+\sqrt{2}}/2 \approx 0.9239$,

$\sin \frac{\pi}{8}=\sqrt{2-\sqrt{2}}/2 \approx 0.3827$.

4th roots of i : angles $\pi/8, 5\pi/8, 9\pi/8, 13\pi/8$:

$0.9239+0.3827i, -0.3827+0.9239i$,

$-0.9239-0.3827i, 0.3827-0.9239i$ (**Check** $\text{first}^2=e^{i\pi/4}$). **Rect $\rightarrow$ polar:** $2+2i$: $r=\sqrt{8}=2\sqrt{2}$, $\theta=\pi/4 \Rightarrow 2\sqrt{2}e^{i\pi/4}$ (**Check**

$2\sqrt{2}(\sqrt{2}/2+i\sqrt{2}/2)=2+2i$).

Geometry: $\times e^{i\theta}$ rotates by θ (lengths mult); $\bar{z}$

mirrors across real axis; unit circle $\{|z|=1\}$

subgroup (closed $|zw|=1$, inv $\bar{z}$). **Plot recipe:**

mark Re on x , Im on y $\rightarrow$ arrow; add tip-to-tail;

mult adds angles, mults lengths.

Groups: $(\mathbb{C}, +)$ $e=0$ inv $-z$; $(\mathbb{C}^*, \cdot)$ $e=1$ inv $\bar{z}/|z|^2$;

$C_4^*=\{\pm 1, \pm i\}$ gen i ; C_8^* gen $(1+i)/\sqrt{2}$; $\{1, -1\}$

group (gen -1); $\{i, -i\}$ **not** ($i \cdot i=-1$ missing).

Mult prop $|z_1 z_2|=|z_1||z_2|$: $z_1=(1+i)/\sqrt{2}$,

$z_2=(1-i)/\sqrt{2}$: $1 \cdot 1=|1|=1$ **Check OK**.

4 — Roots of Unity

Powers $\omega^0.. \omega^{n-1}$ = all n roots cyclically (regular n -gon on unit circle, angle $2\pi/n$ apart); primitive = generator; n distinct roots of $z^n-1=0$.

Primitive: ω_k^n generates all $\iff \gcd(k,n)=1$;

e.g. $n=8$: $k=1, 3, 5, 7$. **Why:**

$\text{ord}(\omega_k^n)=n/\gcd(k,n)$ — shared factors shortcut

the cycle (e.g. $\omega_8^2=\omega_4$, order 4).

Gen test: $g^d \neq 1$ for all proper divisors $d \mid n$ ($\mathbb{Z}_{13}$:

$2^1, 2^2, 2^3, 2^4, 2^6=2, 4, 8, 3, 12 \neq 1$).

Cyclic: $C_n^*=\langle \omega_n \rangle$; reduce first: $\omega_8^2=\omega_4=i$.

Root: $\omega_n=e^{2\pi i/n}$.

Relations (n even; last two $\forall n$): $\omega^n=1$, $\omega^{n/2}=-1$, $\omega^2=\omega_{n/2}$, $\omega^{-1}=\bar{\omega}$, $\omega^{k+n/2}=-\omega^k$, $\omega^{2k}=\omega_{n/2}^k$, $\omega^{n-k}=\bar{\omega}^k$.

Worked $\omega_8=(1+i)/\sqrt{2}$: $\omega_8^4=-1$, $\omega_8^8=1$,

$\omega_8^{-5}=\omega_8^{-5+8}=\omega_8^3=(-1+i)/\sqrt{2}$;

$\omega_{16}^4=e^{8\pi i/16}=i=\omega_4$. $\sqrt{i}=(1+i)/\sqrt{2}$ (**Check**

$((1+i)/\sqrt{2})^2=i$); $\sqrt{-i}=(-1+i)/\sqrt{2}$.

Eval trick: reduce exponents mod n , use $\omega_2=-1$.

Worked $p=x^4-3x^3+x^2-2x-1$ at $\omega_4=i$: powers $\omega_2^4=-1$, $\omega_4^3=-i$, $\omega_4^4=1$.

Then

$1-3(-i)+(-1)-2i-1=1+3i-1-2i-1=-1+i$.

Lecture: $3\omega_4^3-2\omega_4^2+2\omega_4^8=3(-i)+2+2 \cdot 1=5-3i$.

Power drill: $[(\omega_8^3)^4-1]$: $\omega_{12}^8=\omega_4^2=-1$, so

$-1-1=-2$.

5 — Iso $C_n^* \cong \mathbb{Z}_n^+$

Map $\omega_n^k \leftrightarrow [k]_n$; $\times \leftrightarrow +$:

$\omega^k \omega^l=\omega^{k+l} \leftrightarrow [k]+[l]=[k+l]$. Gen $\leftrightarrow$ gen. **Iso:**

bijection preserving op (C_4^* table $\cong \mathbb{Z}_4^+$ table).

Subgroup: subset + closed under op.

Worked $C_4 \leftrightarrow \mathbb{Z}_4$: $[0] \leftrightarrow 1, [1] \leftrightarrow i, [2] \leftrightarrow -1, [3] \leftrightarrow -i$;

$[1]+[2]=[3] \leftrightarrow i \cdot (-1)=-i$;

$[3]+[2]=[5]=[1] \leftrightarrow (-i)(-1)=i$. Tables same shape $\Rightarrow$ iso.

No-iso test: $|G|$ must match + generator orders;

e.g. $|C_2^*|=2$ vs $|\mathbb{Z}_4^+|=4 \Rightarrow$ no iso. **Fill C_5^* via iso:**

$\omega_5^3 \omega_5^4=\omega_5^7=\omega_5^2 \leftrightarrow [3]+[4]=[7]=[2]$. Subgroups:

$C_{2m} \subset C_{2m+1}^* \quad (z^{2m})^2=1 \Rightarrow (z^{2m})^2=1$).

6 — Poly: FTA, Fields, Coeff/Value

Field: $+$, $-$, $\times$, $/$ (no $/0$); $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ yes; $\mathbb{Z}$ no (a/b can leave $\mathbb{Z}$); $\mathbb{N}$ no ($a-b$ can leave $\mathbb{N}$); $\mathbb{Z}_p$ (p prime) yes (every $\neq 0$ invertible). Polys over field K (usually $\mathbb{R}/\mathbb{C}$); some fields give no roots.

Alg closed: every deg ≥ 1 has a root; $\mathbb{C}$ is (FTA); $\mathbb{R}$ is **not** ($x^2+1=0$ no real root).

FTA: deg $n \Rightarrow$ exactly n roots in $\mathbb{C}$ counting mult: $f=a_n \prod (x-x_i)$.

Worked $x^4+16=0$: no real roots; 4 in $\mathbb{C}$; two:

$\sqrt{2}+i\sqrt{2}, -\sqrt{2}+i\sqrt{2}$ (**Check** $(\sqrt{2}+i\sqrt{2})^2=4i$,

$^2=-16$). All four: $\pm\sqrt{2} \pm i\sqrt{2}$ (signs independent);

$|x|=2$, angles $\pi/4, 3\pi/4, 5\pi/4, 7\pi/4$).

Proof (no real roots): for real x , $x^4 \geq 0$ (even power), so $x^4+16 \geq 16 > 0$ — never 0. I.e. $x^4=-16$ impossible over $\mathbb{R}$ ($\text{LHS} \geq 0$). Hence all 4 FTA roots are nonreal. Factor thm:

$f(a)=0 \iff (x-a) \mid f$; $(x-1)^2=x^2-2x+1$

(multiple root); $x^2+1=0$: roots $\pm i$ (none real).

Note: 5 prime in $\mathbb{Z}$ but $5=(2-i)(2+i)$ in $\mathbb{Z}[i]$.

Coeff ($a_0..a_n$); **value:** deg $\leq n$ unique from $n+1$

pts (Vandermonde det $\neq 0$). **Product deg $n+m$**

needs $n+m+1$ pts; pad zeros

$(4+5)x \rightarrow 4+5x+0x^2$. Pointwise

$h(x_i)=f(x_i)g(x_i)$, $O(m)$; schoolbook $O(n^2)$.

Worked $f=1-x, g=2+x$ at $-2, 0, 1$:

$f: 1-(-2)=3, 1, 0$; $g: 2+(-2)=0, 2, 3$;

$h: 0, 2, 0 \Rightarrow h=2-x-x^2$ (**Check** $2+2-4=0, 2, 2-1-1=0$).

Sketch recipe: mark roots $\rightarrow$ lead-coeff sign for ends $\rightarrow y$ -intercept $\rightarrow$ join; graphical mult: sample ≥ 6 x 's, multiply y 's, plot.

Point-count recipe: single deg- $n \rightarrow n+1$ pts; product deg $n+m \rightarrow n+m+1$; FFT $\rightarrow$ next pow2 $\geq$ that.

7 — FFT/IFFT + Costs

Even/odd: $p(x)=p_e(x^2)+x p_o(x^2)$. **Worked**

$1-2x+3x^2-4x^3+5x^4-6x^5+7x^6-8x^7$:

$p_e(y)=1+3y+5y^2+7y^3$, $p_o(y)=-2-4y-6y^2-8y^3$;

depth of recursion tree $=3$ ($8=2^3$).

Butterfly ($k=0..n/2-1$):

$p(\omega^k)=p_e(\omega^{2k})+\omega^k p_o(\omega^{2k})$;

$p(\omega^{k+n/2})=p_e(\omega^{2k})-\omega^k p_o(\omega^{2k})$. Needs $n=2^t$.

Butterfly $n=4$ ($\omega=i$):

$k=0$: $p(1)=p_e(1)+p_o(1)$

$p(-1)=p_e(1)-p_o(1)$

$k=1$: $p(i)=p_e(-1)+i p_o(-1)$

$p(-i)=p_e(-1)-i p_o(-1)$

$N_{\text{FFT}} \geq 2n-1$, smallest pow2 (4+4 coeffs deg3:

product deg6, 7 coeffs $\Rightarrow \text{N}=8$). Pipeline:

coeff $\rightarrow$ FFT $\rightarrow$ vals $\rightarrow$ pointwise $\times \rightarrow$ IFFT $\rightarrow$ coeff.

Zero-pad; extra points OK if same poly. **Idea:** one

(p_e, p_o) pair gives two values (halves work).

IFFT: $a_j = \frac{1}{n} \sum_{k=0}^{n-1} y_k (\omega^{-j})^k$, $j=0..n-1$.

Costs: FFT/IFFT $O(n \log n)$ (depth $\log n \times O(n)/\text{level}$); pointwise $O(n)$; total $O(n \log n)$.

Worked FFT $q=1-x^2+4x^3$ ($N=4, \omega_4=i$):

$q(1)=4$, $q(i)=2-4i$

$q(-1)=-4$, $q(-i)=2+4i$

Butterfly check: $p_e(y)=1-y$, $p_o(y)=4y$;

$q(i)=p_e(-1)+i p_o(-1)=2+i(-4)=2-4i$.

Butterfly table: $k=0$: $p_e(1)=0$,

$p_o(1)=4 \Rightarrow q(1)=0+4=4$, $q(-1)=0-4=-4$.

$k=1$: $p_e(-1)=2$,

$p_o(-1)=-4 \Rightarrow q(i)=2+i(-4)=2-4i$,

$q(-i)=2-i(-4)=2+4i$.

Full pipeline $f=1-x, g=2+x$ ($N=4$):

F : $0, 1-i, 2, 1+i$

G : $3, 2+i, 1, 2-i$

$H=F \cdot G$: $0, 3-i, 2, 3+i$

IFFT: $a_0=(0+3-i+2+3+i)/4=8/4=2$;

$a_1=(-3i-1-2+3i-1)/4=-1$;

$a_2=-1 \Rightarrow h=2-x-x^2$ **Check OK**.

Worked IFFT $y=(-1, i, 1, 2)$: $(\omega^{-j})^k$ ($k=0..3$, $\omega=i$):

$j=0$: $1, 1, 1, 1$ $j=1$: $1, -i, -1, i$

$j=2$: $1, -1, 1, -1$ $j=3$: $1, i, -1, -i$

(digits < 10, no carry needed).
 d decimals $\rightarrow \lceil d/3 \rceil$ base-1000 digits: 120!:
199 $\rightarrow$ 67 (66·3=198<199). **Idea:** every digit must land in $[0, B-1]$ (carry the rest).
GCD monic (lead 1; unique up to $\times \text{const}$):
 $4x^2+12x-7 \rightarrow x^2+3x-7/4$; $\gcd(x^2-1, x-1)=x-1$ ($(x-1)(x+1)$). Euclid via synthetic division.
Worked $f=x^4-8x^3+17x^2+2x-24$,
 $g=x^3-6x^2+11x-6$:
Step 1 (divide, two terms): $x^4/x^3=x$, then $-2x^3/x^3=-2$: $f-(x-2)g=-6x^2+30x-36=-6h$,
 $h=x^2-5x+6$.
Step 2: $g \bmod h$ is 0 $\Rightarrow \gcd(f, g)=h=x^2-5x+6$.
Synthetic $(x^3-6x^2+11x-6)/(x-1)$, coeffs

	1	-6	11	-6	
1, -6, 11, -6:	1		1	-5	6
		1	-5	6	0

Quot x^2-5x+6 , rem 0 (so $x=1$ root). Non-root demo $(x+1)$: 1, -7, 18, -24, rem -24 $\neq$ 0 (so $x=-1$ not a root).
Overdetermined: $h=\gcd$; $\deg h=0 \Rightarrow$ inconsistent ($x-1=0, x+1=0$); $\deg \geq 1 \Rightarrow$ common roots = roots of h . Mini: $x^2-1=0$ (± 1) + $x-1=0$ (1) $\Rightarrow$ common $x=1$. Lecture consistent pair: $x-1=0, 2x-2=0$ (same root 1, $h=x-1$).
Worked: $x^2-11x+30=0$: 25-55+30=0, 36-66+30=0 $\Rightarrow$ 5, 6.
 $x^3-10x^2+27x-18=0$: test 6: 216-360+162-18=0; roots 1, 3, 6 $\Rightarrow$ common **$x=6$** .
Resultant: det condition for common root (eliminates a variable). 2×2 :
 $\text{Res}(x-1, x+1)=(1)(1)-(-1)(1)=2\neq 0 \Rightarrow$ no common root.

9 _____ **Worked Solutions (Step-by-Step)**
_____ **Extra Q&A (Quiz/HW Style)**

FFT size Qs: p with n coeffs needs n points forth+back; FFT needs pow2 so pad.
 $n-1=16 \Rightarrow n=17$ (not pow2) $\Rightarrow$ **cannot use directly**, pad to 32. $(p(x))^2$: $\deg 2(n-1)$, needs $2n-1$ pts $\rightarrow$ next pow2.
IFFT $n=2, y=(1, 1+i)$: $x_0=1=\omega_0^0, x_1=-1=\omega_2^1$;
 $a_0=(y_0+y_1)/2=(2+i)/2=1+0.5i$;
 $a_1=(y_0-y_1)/2=-0.5i$. **Check:** $p(1)=1, p(-1)=1+i$.
Gen a_2 formula ($n=4$): $a_2=(y_0-y_1+y_2-y_3)/4$ ($\omega^{-2k}=1, -1, 1, -1$). **Idea:** IFFT = FFT with ω^{-1} , then $/n$ (**trap:** forgetting $/n$ or the -1).
 C_8^* products ($z_1=1, z_2=-1, z_5=(1+i)/\sqrt{2}, z_6=(-1+i)/\sqrt{2}, z_7=(1-i)/\sqrt{2}, z_8=(-1-i)/\sqrt{2}$):
 $z_2z_5=(-1-i)/\sqrt{2}=z_8$; $z_5z_7=1=z_1$; $z_6z_8=1=z_1$.
Inverse of $(-1-i)/\sqrt{2}$: $|z|^2=1$, inv $=\bar{z}=(-1+i)/\sqrt{2}=z_6$.
Group tests: $2a+7bi$ ($a, b\in\mathbb{Z}$) additive: closed, $e=0$, inv $-z \Rightarrow$ **group**. $3a+3bi, ab\neq 0$ multiplicative: $1=3a+3bi \Rightarrow a=1/3\notin\mathbb{Z} \Rightarrow$ **not** (no identity).
FFT $n=32$: depth 5 ($2^5=32$); butterfly pairs $(\omega^0, \omega^{16}), (\omega^1, \omega^{17})$; even/odd parts $\deg 15$, split again $\rightarrow 4$ polys $(p_{ee}, p_{eo}, p_{oe}, p_{oo})$. C_n^+ = cyclic additive order n ($\cong \mathbb{Z}_n^+$), gens $[k]$ with $\gcd(k, n)=1$; chain $C_{2^m}^* \subset C_{2^{m+1}}^*$.
 $p(\omega^0)=p_e(1)+p_o(1)$
 $p(\omega^{16})=p_e(1)-p_o(1)$
 $p(\omega^1)=p_e(\omega^2)+\omega p_o(\omega^2)$
 $p(\omega^{17})=p_e(\omega^2)-\omega p_o(\omega^2)$
Inconsistent check: $\gcd(x-1, x+1)=1$ ($\deg 0$) $\Rightarrow$ **inconsistent** $((x+1)-(x-1)=2\neq 0)$.

_____ **Mod + Groups/FLT Drills**
Reduce first:
 $[125]_{17}+[241]_{17}=[366]_{17}$; $366-21\cdot 17=366-357=9$.
 $[391]_{21}\cdot [457]_{21}$: $391-18\cdot 21=13, 457-21\cdot 21=16$;
 $13\cdot 16=208-9\cdot 21=19$.
 $[94]_7$: $94-13\cdot 7=94-91=3$.
 $([157]_{13}+[249]_{13})\cdot [83]_{13}$: $157-12\cdot 13=1,$

$249-18\cdot 13=249-234=15$; $15-13=2$; $1+2=3$;
 $83-6\cdot 13=5$; $3\cdot 5=15-13=2$.
HW08 reduce-first:
 $[123]_{17}+[245]_{17}=11$ ($368-357$). $[391]_{23}\cdot [457]_{23}=0$ ($391=17\cdot 23$).
 $([156]_{11}+[247]_{11})\cdot [83]_{11}$: $2+5=7, \times 6=42-33=9$.
 $[312]_{15}\cdot ([184]_{15}+[257]_{15})$: $12\cdot (4+2)=72-60=12$.
Find x :
 $143 \equiv x \pmod{13}$: $143=11\cdot 13 \Rightarrow 0$
 $12345 \bmod 11$: $11\cdot 1122=12342 \Rightarrow 3$
 $54321 \bmod 7$: $7\cdot 7760=54320 \Rightarrow 1$
 $147 \bmod 13$: $13\cdot 11=143 \Rightarrow 4$
Find n :
 $24 \equiv 0 \pmod{n} \Rightarrow n|24$: 1, 2, 3, 4, 6, 8, 12, 24.
 $15 \equiv 1 \pmod{n} \Rightarrow n|14$: 1, 2, 7, 14.
 $124 \equiv 4 \pmod{n} \Rightarrow n|120, n>4$:
5, 6, 8, 10, 12, 15, 20, 24, 30, 40, 60, 120. $18 \equiv 0, 0<n\leq 18$: 1, 2, 3, 6, 9, 18.
Classes: mod 7: 7 classes $[0]..[6]$.
 $[2]_5=\{\dots, -8, -3, 2, 7, 12, \dots\}$; $[4]_7=\{\dots, -3, 4, 11, \dots\}$;
 $[-3]_8=[5]_8=\{\dots, -3, 5, 13, \dots\}$. $\{1, -1\} \subset C_4^*$ subgroup (closed, $e=1$, self-inv).
Notation $\leftrightarrow$:
 $[8]_{11}+[9]_{11}=[6]_{11} \iff 8+9 \equiv 6 \pmod{11}$
 $(17-11)=6$; $[4]_7\cdot [5]_7=[6]_7 \iff 4\cdot 5 \equiv 6 \pmod{7}$ ($20-14=6$).
 $17+9 \equiv 2 \pmod{8} \iff [17]_8+[9]_8=[2]_8$
 $(26-24)=2$; $6\cdot 8 \equiv 3 \pmod{5} \iff [6]_5\cdot [8]_5=[3]_5$ ($48-45=3$).
Groups: $(\mathbb{Q}, +)$ group; $(\mathbb{Q}, \cdot)$ **not** (0 no inverse).
 $(\mathbb{Z}_4, +)$ group ($e=[0]$, inv $[-a]$). $\mathbb{Z}_4^*=\{1, 2, 3\}$ **not** ($2\cdot 2=0$).
Mult groups — which n admit one? $n=3, 5, 13, 17$ yes ($|G|=2, 4, 12, 16$); 10, 15 no. Find gens: $\mathbb{Z}_5^*$ gens $\{2, 3\}$ (ord=4).
Possible orders (divisors of $|G|$): $2\rightarrow\{1, 2\}$;
 $4\rightarrow\{1, 2, 4\}$; $12\rightarrow\{1, 2, 3, 4, 6, 12\}$;
 $16\rightarrow\{1, 2, 4, 8, 16\}$. $|\mathbb{Z}_{24}|=\varphi(24)=8$ (odd, not mult of 3: 1, 5, 7, 11, 13, 17, 19, 23); x odd (e.g. 3, 5, 7) never gives $[a]^x=[1]$ so every even x works).
Gen of $\mathbb{Z}_{13}^*$: $|G|=12$, orders $\in\{1, 2, 3, 4, 6, 12\}$;
 $2^2=4, 2^3=8, 2^4=3, 2^6=12\neq 1 \Rightarrow \text{ord}(2)=12 \Rightarrow$ generator.
FLT big powers: $19^6 \bmod 7$ OK ($7\nmid 19$); 21^6 **no** ($\gcd=7$); $11^8 \bmod 9$ **no** (9 composite).
 19^{68} : $19 \equiv 5, 68=6\cdot 11+2$ ($66+2$), $5^2=25 \equiv 4$.
 $271^{24} \bmod 11$: $271-24\cdot 11=271-264=7$.
 $24=10\cdot 2+4, 7^4=2401-218\cdot 11=2401-2398=3$.
 $290^{293} \bmod 23$: $290-12\cdot 23=290-276=14$,
 $293=22\cdot 13+7$ ($286+7$), $14^7 \equiv 19$.
Simplify without FLT: $[16^4]_7=[2]$; $[10^{100}]_9=[1]$;
 $[23^2]_9=[7]$ ($23 \equiv 5, 25-18$); $[42]_{12}=[6]$.
More Euler: $5^{100} \bmod 13$: $\varphi=12$; $100=12\cdot 8+4$;
 $5^4=625-48\cdot 13=625-624=1$. $2^{1000} \bmod 7$:
 $1000=6\cdot 166+4$; $2^4=16 \equiv 2$.
Compute φ : $\varphi(8)=8-4=4$;
 $\varphi(24)=\varphi(8)\varphi(3)=4\cdot 2=8$;
 $\varphi(100)=\varphi(4)\varphi(25)=2\cdot 20=40$. **Trap:**
 $\varphi(4)\varphi(2)=2\cdot 1=2\neq 4=\varphi(8)$ ($\gcd(4, 2)\neq 1$).
Euler trap: $2^4 \bmod 8=0\neq 1$ though $\varphi(8)=4$ ($\gcd\neq 1$).
Mod x^3-1 (Quiz4 Q6): $x^3 \equiv 1$:
 $[2x^5-9x^3+5x+9]=2x^2-9+5x+9=[2x^2+5x]$.
Square after reducing:
 $[(2x^2+5x)^2]=[4x^4+20x^3+25x^2]$; $x^4 \equiv x$,
 $x^3 \equiv 1 \Rightarrow [25x^2+4x+20]$.

_____ **Complex + FFT Drills**

Sums (plot tip-to-tail, check):
 $(-2+4i)+(-3-7i)=-5-3i$
 $(-2-i)+(2+i)=0$
 $(1-2i)+(1+2i)=2$
 $|z|^2=z\bar{z}$: $(-2-3i)$: $4+9=13$; $4i$: 16;
 $\bar{z}=4i \Rightarrow z=-4i, |z|^2=16$.
Div: $(2+6i)/2=1+3i$.
 $(3+4i)/(1-6i)=(3+4i)(1+6i)/37=(-21+22i)/37$.

$\approx -0.5676+0.5946i$ **Check** $(37\cdot(-0.5676))=-21, 37\cdot 0.5946=22)$. **vs dot:** $z_1z_2=(ac-bd)+(ad+bc)i$ vs $u\cdot v=ac+bd$; $|z_1z_2|^2=(a^2+b^2)(c^2+d^2)$ vs $(u\cdot v)^2=(ac+bd)^2$; coincide $\iff ad=bc$.
E.g. $z_1=1+2i, z_2=3+4i$:
 $|z_1z_2|^2=(1+4)(9+16)=125$ vs $(3+8)^2=121$ ($1\cdot 4\neq 2\cdot 3$).
Even/odd twice of $p_e=1+3y+5y^2+7y^3$,
 $p_o=-2-4y-6y^2-8y^3$:
 $p_{ee}(z)=1+5z, \quad p_{eo}(z)=3+7z$
 $p_{oe}(z)=-2-6z, \quad p_{oo}(z)=-4-8z$
Big-O tree (n leaves, balanced binary): depth $\log_2 n$; bottom-only work $\Rightarrow O(n)$; $O(n)$ per level $\Rightarrow O(n \log n)$.
 C_5^* row via iso:
 $\omega_5^4\cdot\omega_5^4=\omega_5^8=\omega_5^3 \leftrightarrow [4]+[4]=[8]=[3]$. Gen $\omega_5 \leftrightarrow [1]_5$ powers fill table.

_____ **Tables + FFT Drills**

Mod-8 $\times$ table (nonzero rows): Rule: row a has 1 $\iff a$ invertible; row a hits 0 $\Rightarrow$ zero divisor. Units = odd $\{1, 3, 5, 7\}$ ($\gcd=1$); $1^{-1}=1, 3^{-1}=3$ ($9-8=1$), $5^{-1}=5$ ($25-24=1$), $7^{-1}=7$ ($49-48=1$). Even rows lack 1. **Zero divisors:** $2\cdot 4=8 \equiv 0, 6\cdot 4=24 \equiv 0$ ($24-24$).
FFT $f=2+3x+5x^3$ ($N=4, \omega_4=i; p_e(y)=2, p_o(y)=3+5y$):
 $f(1)=2+3+5=10$; $f(i)=2+3i+5(-i)=2-2i$.
 $f(-1)=2-3-5=-6$; $f(-i)=2-3i+5i=2+2i$.
Butterfly table (f): $k=0$: $p_e(1)=2, p_o(1)=8 \Rightarrow f(1)=2+8=10, f(-1)=2-8=-6$.
 $k=1$: $p_e(-1)=2, p_o(-1)=-2 \Rightarrow f(i)=2+i(-2)=2-2i, f(-i)=2-i(-2)=2+2i$.
DeMoivre power:
 $((1+i)/\sqrt{2})^8=(e^{i\pi/4})^8=e^{2\pi i}=1$ (angle $\times 8$).
 $((1-i)/\sqrt{2})^2=-i, ((1-i)/\sqrt{2})^4=-1$ (angle $\times 4=-\pi$). **Decrypt proof:** $m \equiv c^d=(m^b)^d=m^{bd}=m^{1+k(p-1)}=m\cdot(m^{p-1})^k \equiv m$ (FLT, p prime, $p\nmid m$).

_____ **Tables + Step-by-Step Recoveries**

Mod-8 $\times$ table (read inverses/zero divisors; odd rows permute, even rows hit 0):

$\times$	1	2	3	4	5	6	7
1	1	2	3	4	5	6	7
2	2	4	6	0	2	4	6
3	3	6	1	4	7	2	5
4	4	0	4	0	4	0	4
5	5	2	7	4	1	6	3
6	6	4	2	0	6	4	2
7	7	6	5	4	3	2	1

C_4^* vs $\mathbb{Z}_4^+$ (same shape $\Rightarrow$ iso $\omega_4^k \leftrightarrow [k]_4$):

*	1	i	-1	$-i$	+	0	1	2	3
1	1	i	-1	$-i$	0	0	1	2	3
i	i	-1	$-i$	1	1	1	2	3	0
-1	-1	$-i$	1	i	2	2	3	0	1
$-i$	$-i$	1	i	-1	3	3	0	1	2

C_5^* table via iso ($\omega=\omega_5$; $\omega^i\cdot\omega^j=\omega^{(i+j) \bmod 5}$):

*	1	ω	ω^2	ω^3	ω^4
1	1	ω	ω^2	ω^3	ω^4
ω	ω	ω^2	ω^3	ω^4	1
ω^2	ω^2	ω^3	ω^4	1	ω
ω^3	ω^3	ω^4	1	ω	ω^2
ω^4	ω^4	1	ω	ω^2	ω^3

Recover $h=a+bx+cx^2$ from values (0, 2, 0) at $(-2, 0, 1)$: $a-2b+4c=0, a=2, a+b+c=0$.
Step 1: $a=2$. **Step 2:** $b+c=-2, -2b+4c=-2 \Rightarrow b-2c=1$.
Step 3: subtract: $3c=-3 \Rightarrow c=-1, b=-1$. **Box:** **$h=2-x-x^2$** . **Check:** $-2+2-4=0, 2, 2-1-1=0$.
Why unique: Vandermonde det at $-2, 0, 1$:
 $(0+2)(1+2)(1-0)=2\cdot 3\cdot 1=6\neq 0$. **Read graph $\rightarrow$ number:** point $(x, y) \rightarrow x+yi$; length r , angle $\theta \rightarrow r(\cos \theta + i \sin \theta)$.